Q1)

1. مش علينا
2. F , Feature Detector
3. مش علينا
4. F, Image rotation
5. F, Same label
6. F
7. مش علينا
8. T
9. T
10. F, False positive
11. T
12. T

    $$IoU = \frac{Intersection}{Union} = \frac{1 \times 1}{(2 \times 1) + (2 \times 4) - (1 \times 1)} = \frac{1}{9}$$

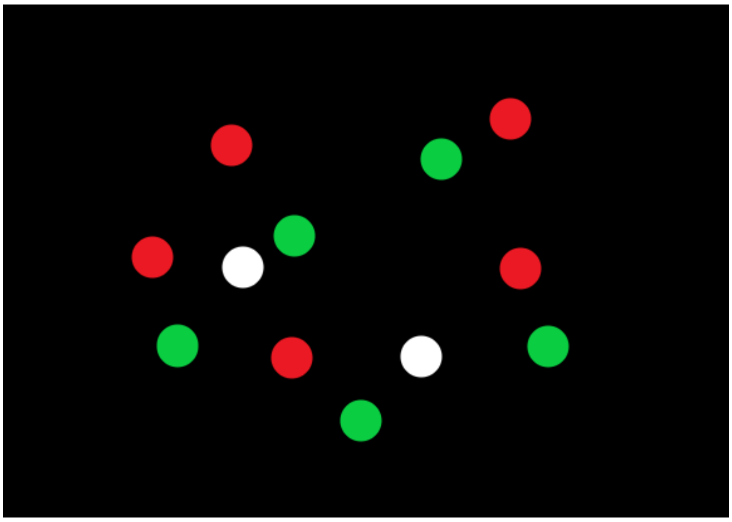
    Note:

    عشان منحسبش الintersection مرتين $-(1 \times 1)$

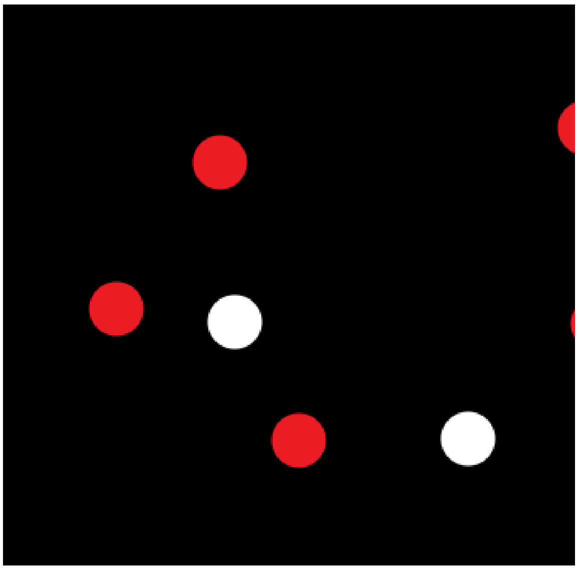
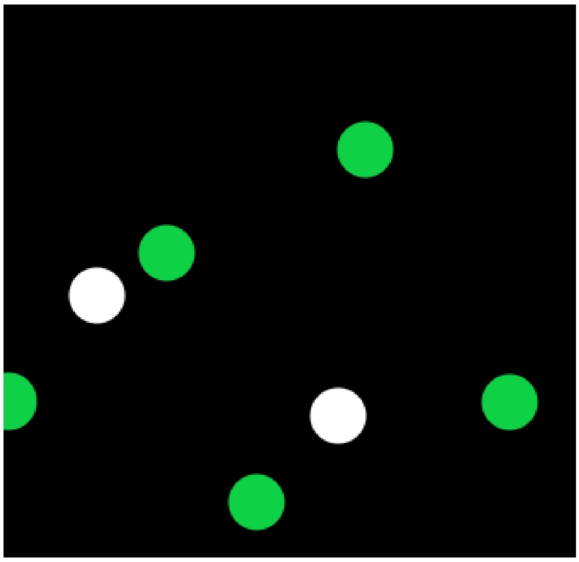
13. مش علينا
14. T

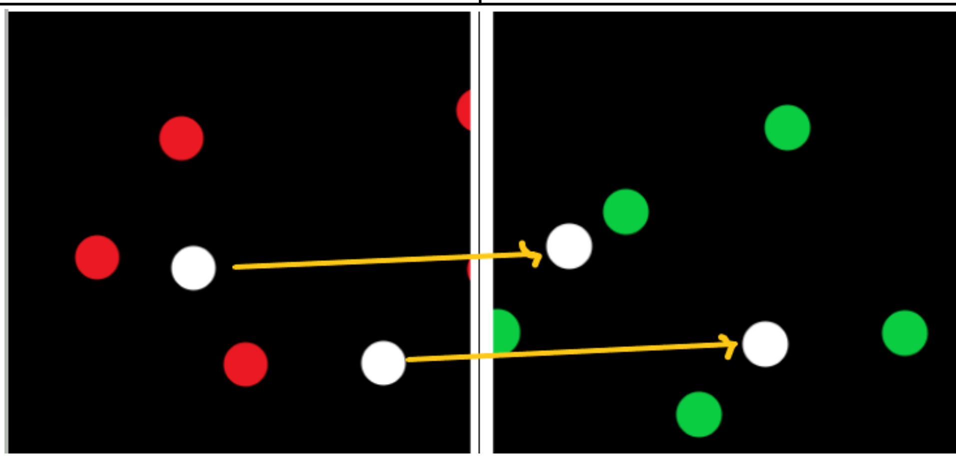
Q2)

A)



let C_R be the camera that sees Red and C_G be the Camera that sees Green

C_R view:	C_G view:
	



i.

C_R : sees 5 red points and 2 White points

C_G : sees 5 green points and 2 White points

ii.

Yes, Since we have 2 points which are visible to both cameras (white points) so we can get 2 matches which is sufficient for estimating a homography matrix of translation that contains only 2 unknowns

iii.

The RANSAC algorithm

Algorithm:

1. **Sample** (randomly) the number of points required to fit the model.
2. **Solve** for model parameters using sample.
3. **Score** by the fraction of ***inliers*** within a preset threshold of the model.
4. **Repeat** 1-3 until the best model is found with high confidence.

Parameters:

- Initial number of points s
  - Typically minimum number needed to fit the model
- Distance threshold T
  - Choose T so probability for inlier is high (e.g. 0.95)
- Number of iterations N
  - Choose N so that, with probability p, at least one random sample set is free from outliers (e.g. p = 0.99)

B)

i.

- Find features (color, gradients, texture, etc.).
- Initialize windows at individual feature points.
- Perform mean shift algorithm for each window until convergence.
- Merge windows that end up near the same "peak" or mode.

Mean Shift Algorithm

1. Choose a search window size.
2. Choose the initial location of the search window.
3. Compute the mean location (centroid of the data) in the search window.
4. Center the search window at the mean location computed in Step 3 (shift)
5. Repeat Steps 3 and 4 until convergence.

Parameters:

- Windows size

ii.

• Very large window size is **fast but not accurate** and can lead to mistakenly segmenting of multiple objects as one object
• small window size is **slower but more accurate** but very small size can lead to noise being detected as an object

iii.

1. Doesn't assume a shape of the clusters (K-means assumes spherical clusters)
2. Not Sensitive to outliers
3. Find variable number of clusters (no prior knowledge of the count of objects is needed)

Q3)

A)

1. مش علينا
2. مش علينا
3. مش علينا
4. Because it uses 2000 ConvNets for feature extraction from the RoIs which requires 2000 forward passes of each image

5. Fast R-CNN, By first extracting features using 1 ConvNet then perform region proposal on the extracted features and by replacing the SVM as the classifier with a softmax

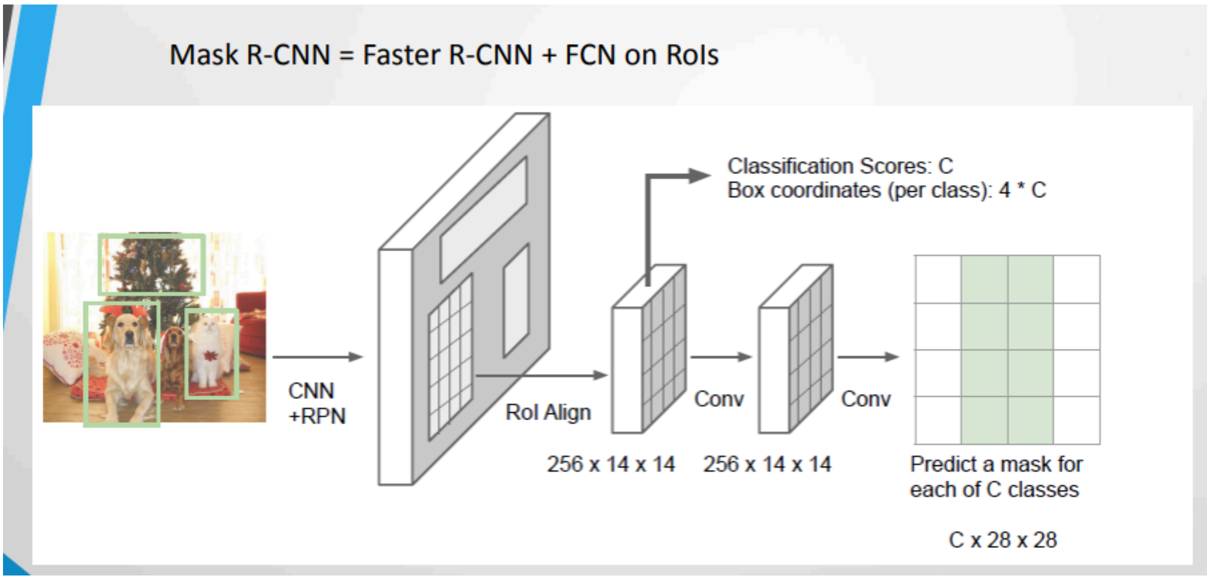
6. The RPN is used to learn proposing ROIs instead of using selective search

7. $Output = S \times S \times (5 \times B + C)$, where:
    S : The grid size to divide the image onto
    5: $dx, dy, dh, dw,\ confidence$
    B: The number of bounding boxes around each anchor
    C: The number of classes to predict from

8. $Output = (S \times S \times (B \times (5 + C))$, where:
    Each anchor box has it's class probabilities

B)

Mask R-CNN:

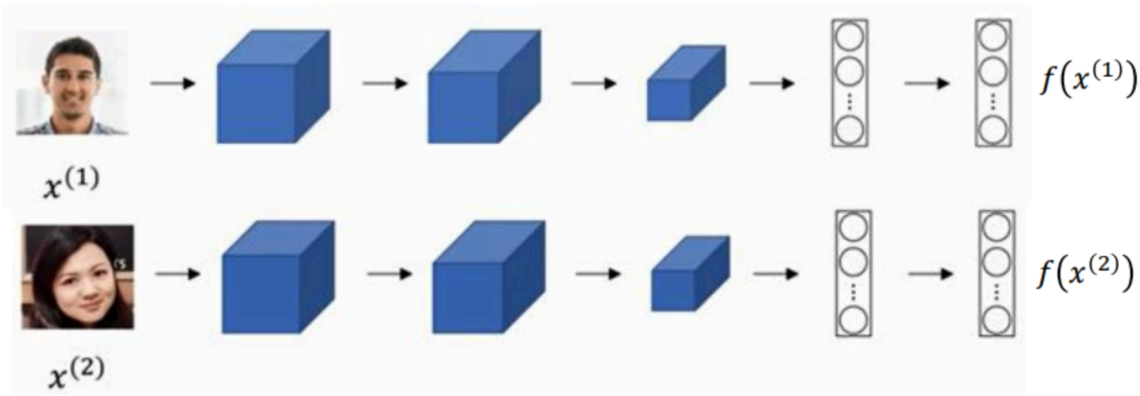


Q4)

A)

Architecture:

Siamese Network



$$d(x^{(1)}, x^{(2)}) = \|f(x^{(1)}) - f(x^{(2)})\|_2^{\,2}$$

If $x^{(i)}, x^{(j)}$ are the same person, $\|f(x^{(i)}) - f(x^{(j)})\|^2$ is small.

If $x^{(i)}, x^{(j)}$ are different persons, $\|f(x^{(i)}) - f(x^{(j)})\|^2$ is large.

(Lec 8, Slide 48)

Learning Procedure:

1) **Input:** Triplets of an anchor image (training image), a positive image *(image from the same class as anchor)* and a negative image *(image from a different class than the anchor)*
2) **Learning Objective:** Minimize distance between anchor and positive images and maximize distance between anchor and negative images
3) **Loss Function:** Triplet loss

$$L(A, P, N) = \max(\|f(A) - f(P)\|_2^{\,2} - \|f(A) - f(N)\|_2^{\,2} + \alpha, 0)$$

B)

1) From all the photos (including your photo):
    a) Extract the features of the photo using SIFT
    b) Learn “visual vocabulary”
    c) Quantize features using visual vocabulary
    d) Represent photos by frequencies of “visual words”
    e) Convert each histogram into a normalized feature vector
2) Use a k-NN model to list the top images matching your template image based on a similarity(distance) metric
3) If the image distance is less than a certain threshold:
    a) Add the photo to the subset of images which you appear in